

“NATURE-INSPIRED ADAS ALGOS Will Be MUCH BETTER Than Robotic Paradigm Algos”

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Most of the ADAS-enabled vehicles of today function on algorithms (algos) inspired by robots used in a logistics setup. Can ADAS algos inspired by the human brain and nature be better? In a conversation with EFY’s Mukul Yudhveer Singh, Gagandeep Reehal, Co-Founder, CEO & CTO, Minus Zero, illustrates this using the ‘sunset mistaken for red light’ example, shedding light on where India’s Minus Zero is headed!

What is Minus Zero?

Minus Zero is pioneering a revolutionary shift in the realm of advanced driver assistance systems (ADAS), transitioning from a robotics-centric to an AI-centric approach. This sets us apart from other players in the ADAS arena. Our focus lies in developing a nature inspired AI (NIA) software for the autonomous vehicles that surpasses the current ADAS technology. This model would be equipped with the intrinsic knowledge of basic functioning of the world that current systems lack. In simpler terms, we’re laying the groundwork for artificial intelligence to drive autonomously, capable of making decisions independently.

What will this generalised AI for autonomous driving accomplish?

Initially, our aim is to offer this advanced AI as a platform to automotive original equipment manufacturers (OEMs). This platform will empower them to integrate Tesla-like autonomous capabilities into their vehicles, regardless of geographies, vehicle form factor, or unstructured roads (with lane markings or without).

Are not many companies pursuing similar goals?

Most of the autonomous vehicle startups we have today are focused more on a robotics-based foundation, akin to technologies found in warehouse robots. However, our distinct

AI-first approach is rare. Recently, a handful of companies have started to work in the field of AI-driven autonomous vehicles.

Could you elaborate on your ‘differentiated approach’?

To understand our approach, it’s essential to trace back to the origins of this industry, which can be pinpointed to the Darpa Challenge in the 2000s. Traditionally, most efforts have focused on a robotics-first strategy, emphasising obstacle detection and mapping the environment. While effective in predictable environments, this approach falters in unfamiliar or unmapped terrains, leading to accidents. Our AI-first approach represents a paradigm shift. Rather than